Local bounds and attainable common grades

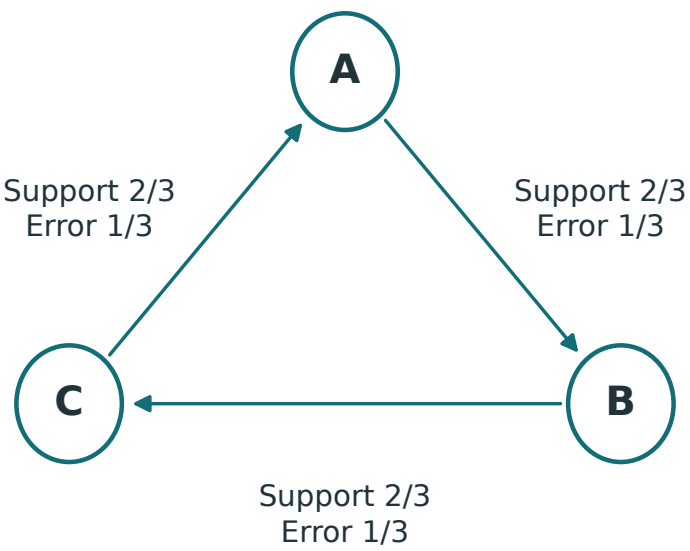
Known criteria; w = (1/3, 1/3, 1/3).
$\tau = 9/20$, $M = 3$. No latent criterion ties.

(a) Known criteria produce a cycle

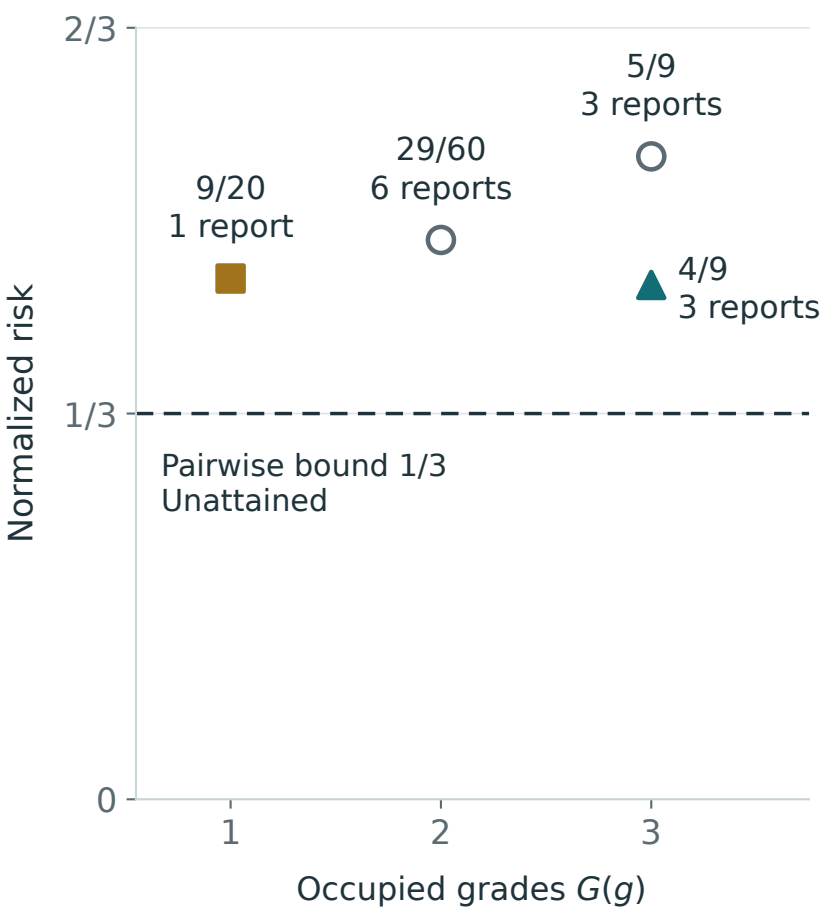
Criterion 1: A > B > C

Criterion 2: B > C > A

Criterion 3: C > A > B



(b) All 13 coherent reports



At most 2 grades
Unique optimum: all-withhold
Risk 9/20

At most 3 grades
Three co-optimal strict reports: ABC, BCA, CAB
Risk 4/9; improvement 1/180
(orders are listed from highest to lowest)

Exactly two grades: risk 29/60 for all six reports.
All-withhold is excluded.